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Clearer, crisper, and more lifelike: adjectives we most often encounter when High Definition (HD) video content is being talked about. So what exactly is HD? According to the Advanced Television Systems Committee, the group that developed the industry standards for digital TV in the US, it is the broadcasting of digital content at resolutions as great as 1280 x 720 or 1920 x 1080, which is higher than that of traditional television systems like NTSC (720 x 480) and PAL (768 x 576). The higher resolution means the picture captures more detail, and so is crisper and more vibrant. HDTV is steadily gaining mainstream acceptance in many countries—in the US, most TV Studios have an HD channel to their name, and in the UK, there's Sky HD and BBC HD. Here in India, things are just starting to hot up. The introduction of new players in the Direct To Home (DTH) service space—like Reliance and Bharti Telemedia—has renewed talk of HDTV channels being introduced here. Is that talk to be believed?

There is quite a lot of uncertainty surrounding the



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providers and content broadcasters. The content providers have to migrate to new equipment that allows them to record and broadcast HD content, and also receive it on their ground stations, and the content broadcasters need to upgrade their infrastructure too. So where does the difficulty lie?

The biggest obstacle for the delivery of HD content is that production houses need to change to equipment which can record at higher resolutions. First, storage capacity needs to be scaled up; and second, HD content also requires a lot of bandwidth for broadcasting—something that is scarce in India. The bigger companies like Reliance ADAG and Bharti Telemedia have set up optical fibre networks across the country. This should address the bandwidth problem for end users, and in the future pave the way for triple-play services: videoconferencing, VoD, and VoIP. The successful launch of India's Insat-4B satellite is also helping to ease the transponder congestion that Indian channels currently face. It currently supports the beaming of Prasar Bharati's DD Direct, a freely available (or "free-to-

The Homecoming Of HD

We have heard too much for too long about High Definition Television in India—let's take a look at the ground realities

delivery of HD content in India for the next two years: there is ambiguity on how the broadcasters will eventually deliver content.

The Current Setup

Our first experience of network television was thanks to the local *cablewala*. When the DTH operators stepped in, they promised to bring with them digital TV clarity—which would be free from snow and noise, and having better contrast ratios. They also promised ancillary services like VoIP, Video on Demand, and Interactive TV. Even after a year, most of these have remained just jargon on paper—for the common man at least.

The successful entry of HDTV content in India depends on two key entities—the content

air") DTH service across India, and will include more players soon.

The Eleventh Five-Year-Plan has mooted the allocation Rs 5,000 crore based on the recommendations of a sub-group comprising private and public broadcasters called "Going Digital" to roll out HD content across India in an eleven-stage process starting with Delhi (2008-09) (and extending to other metros, and eventually other cities and towns). The aim is to cover and broadcast the 2010 Commonwealth Games in HD.

The Compression Issue

HDTV requires compression during broadcast and reception; the problem here is that it employs either of two alternative forms of

compression—MPEG-2 and MPEG-4 AV. So the Set-Top Box (STB) will have to support both technologies unless HD broadcast standardisation takes place. Providers currently supply STBs that support only one of the two compression technologies.

The Hardware

Moving to HD will mean that most households in India will have to change their TV sets. The hardware has, as you can see too well, overstepped the content. Walk into any TV showroom and you will find a range of HD-capable TVs and video players, with there being hardly—if any—HD content available.

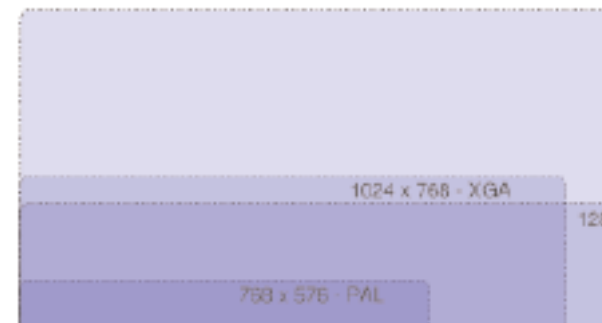
On the content provider's side, the HD equipment is expensive, and requires an overhaul of enterprise-wise workflow systems—an exercise not all content production studios are

willing to go through just yet. Still, new production houses like NDTV already have HD-ready infrastructure like cameras and video switchers. DishTV and Network 18 are also in the process of making their systems HD-capable, and plan to provide HD content when it is required. According to Piyush Gupta, the Head of Operations at Network 18, "We are currently using SD equipment, which will be phased out to make the facility HD compatible. We will achieve this gradually, starting from replacing equipment used for content acquisition right down to the studio and production chain"

The prices of HDTVs are also falling every month; buoyed by this, Indian movie studios are actively considering the distribution of HD content on dual-layered DVDs, a recent example being the distribution deal of HD-format Bollywood movies between New Medium Enterprises and leading Bollywood studios in January 2007. Ten titles and an HD player capable of playing HD-VMD titles are available for \$299 (Rs 12,000) through Indian retailers and at various Indian e-commerce sites.

The Rollout

Everyone we spoke to was optimistic about the future of HDTV in India; however, they all agree that HD content will not be available for at least two more years. During the initial entry years of HD, the normal tradition worldwide had been that the Standard



An illustration of the resolutions at which HD TV is broadcast compared to the regular broadcast system



Standard definition TV in India is 'good enough' according to most Indian consumers. Currently, there is limited willingness to pay for the incremental improvement offered by HDTV"

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